

Perigo Mine Site — Geologic, Mineralogic & Geochemical Interpretation

Interpretation of every table and figure produced by `Perigo_Mine_EDA.ipynb`

Regional and site geologic setting

The data come from a real USGS data release (Manning, Runkel, Ball, McGee, Romero-Matos, Muñoz-Sepúlveda, 2026, *Geochemistry and Environmental Tracer Data for Groundwater and Stream Water from the Perigo Mine Site, Colorado, 2024*, <https://doi.org/10.5066/P143OCZP>). USGS background on the release describes it as stream-water and groundwater samples collected July–October 2024 at the Perigo Mine site in Gilpin County, northern Colorado, as part of a study characterizing the local groundwater flow system in the hillslope that hosts the mine's draining crosscut tunnel and identifying sources of contamination reaching the stream [USGS Colorado Water Science Center]. Related USGS project notes describe fieldwork that included a step-injection aquifer test while drilling a bedrock borehole at the site in September 2023 [USGS, "Persistent Science Challenges in Legacy Mine Land Site Management"], which corresponds to the nested bedrock-well network (P5, P8, P9A/B/C) seen in the site map. Gilpin County sits squarely in the historic **Central City–Idaho Springs mining district**, part of the Colorado Mineral Belt — a NE-trending belt of Laramide-age porphyritic intrusions that cut Precambrian metamorphic and granitic basement rock and generated polymetallic sulfide veins (pyrite, chalcopyrite, sphalerite, galena, with quartz/carbonate gangue and locally telluride-hosted Au/Ag). The "Perigo Mine" itself is a legacy underground gold/base-metal working; its draining adit (the "Portal Drainage" site) is the classic geochemical engine of the whole dataset — sulfide minerals exposed to oxygenated water inside the abandoned workings generate **acid rock drainage (ARD)**, and every downstream table and plot in the notebook is essentially tracking that signal through the hillslope and into the stream.

Section 2 — Site map (cell 6)

The map shows two spatially distinct clusters: a tight nest of bedrock monitoring wells (P5, P8, P9A–C) near the mine hillslope, and a long, roughly linear string of stream, streambank-well, and spring sites (STR1→GG-0250→...→GG-1311/1038) running ~1.3 km down-valley. That linear string is the **synoptic sampling reach** — a standard USGS tracer-dilution/mass-loading survey design (Kimball/Runkel-style) used to walk downstream and see exactly where contaminated groundwater enters the channel. The Portal Drainage point (PTL1) sits geographically between the bedrock-well nest and the head of the stream reach — consistent with it being the adit that drains the mine workings into the valley.

Section 3 — Water-chemistry summary statistics (cells 8–9)

The per-site-type medians are the clearest single geochemical fingerprint in the dataset:

Portal Drainage: pH $\approx$ 3.7, SpC $\approx$ 645 μ S/cm, SO₄ $\approx$ 333 mg/L, **negative** alkalinity (–6 mg/L as HCO₃).

Negative/near-zero alkalinity means the water has no acid-neutralizing capacity left — textbook ARD produced by pyrite (FeS₂) oxidation: $\text{FeS}_2 + \text{H}_2\text{O} + \text{O}_2 \rightarrow \text{Fe}^{2+}/\text{Fe}^{3+} + \text{SO}_4^{2-} + \text{H}^+$.

LBi vs RBI (left-bank inflow / right-bank inflow): these aren't grab samples — they're inferred end-member chemistries back-calculated from the mass-balance/tracer-dilution method, which is why many field

parameters are blank for them. Geochemically, LBI is far more acidic ($\text{pH} \approx 6.1$) and SO_4 -rich ($\approx 233 \text{ mg/L}$) than RBI ($\text{pH} \approx 7.0$, $\text{SO}_4 \approx 10.6 \text{ mg/L}$) — meaning the mine-impacted groundwater is entering the stream preferentially from one bank (the side the adit hillslope drains toward), not diffusely from both.

Section 4 — Major-ion boxplots by site type (cell 11)

Portal Drainage and LBI stand well apart from Bedrock Wells, Springs, and RBI on all three panels: highest SpC, lowest pH, highest SO_4 . This is the same ARD/left-bank signal from Section 3, just visualized — it confirms the acidic, sulfate-rich, mine-influenced water is a distinct hydrochemical population, not just the tail of natural variability.

Section 5 — Trace-metal boxplots (cell 13)

Zn, Cu, Cd, Pb, Mn, and Al (log-scaled) are all 2–3 orders of magnitude higher in Portal Drainage and LBI than in Bedrock Wells, Springs, or RBI. This is the mineralogical signature of the vein itself: under the low pH generated by pyrite oxidation, the other sulfides in the vein assemblage go into solution —

Sphalerite (ZnS) → dissolved Zn (highest concentrations of the suite, consistent with sphalerite typically being volumetrically dominant in these polymetallic veins)

Chalcopyrite (CuFeS_2) → Cu (and additional Fe)

Galena (PbS) → Pb, though Pb mobility is usually capped by formation of insoluble anglesite (PbSO_4) or cerussite, which is why Pb concentrations, while elevated, don't reach Zn-like levels

Cd rides along as a trace substituent in sphalerite's crystal lattice — its concentration pattern almost always mirrors Zn's

Mn and Al come partly from sulfide-hosted trace Mn and partly from acid attack on host-rock silicates/clays, which only mobilize Al efficiently once pH drops below ~ 5

Section 6 — Correlation heatmap (cell 15)

The block of strong positive correlations linking SpC, SO_4 , and all six metals reflects one shared process (sulfide oxidation/ARD generation), not six independent ones. The strong **negative** correlation between pH and that same block is the direct chemical control: lower pH keeps metal-hydroxide solubility products from being exceeded, so metals stay dissolved. Alkalinity correlating positively with pH and negatively with the metal block is the flip side — alkalinity is the water's remaining buffering capacity, and once it's consumed, pH crashes and metals mobilize. NO_3 shows only weak correlations across the board, telling you this system's chemistry is driven by hard-rock sulfide weathering, not agricultural/septic nitrogen inputs.

Section 7 — Slug-test conductivity time series (cell 17)

This is a hydrologic (not mineralogic) method — background-corrected specific conductance (BCSC) breakthrough curves from salt-slug streamflow measurements at each station, used purely to compute discharge for the load calculations later. Narrower, taller peaks indicate lower streamflow (less dilution of the salt slug); broader, lower peaks indicate higher flow.

Section 8 — Streamflow & Zn-load profile downstream (cell 19)

The sharp co-located step in both streamflow and Zn load between roughly 300–420 m downstream is the notebook's central diagnostic: it pinpoints a discrete zone where ARD-impacted groundwater — most plausibly water that has passed through the mine workings and hillslope — enters the channel. This is the

same feature that later shows up as station **GG-0423** dominating the per-reach contribution table (Section 11).

Section 9 — Piper diagram (cell 23)

Most samples cluster on the Ca-Mg side of the cation triangle (low Na+K), with the anion composition spreading between HCO₃-dominated and SO₄/Cl-dominated end members — i.e., waters range from **Ca-Mg-HCO₃ type** (background groundwater recharged through fractured silicate/minor-carbonate bedrock) to **Ca-Mg-SO₄ type** (ARD-impacted water, sulfate supplied by pyrite/base-metal-sulfide oxidation). Bedrock wells and Portal-influenced sites trend toward the SO₄ corner; springs and RBI trend toward HCO₃ — consistent with springs representing shorter, less mine-influenced flow paths and bedrock wells intersecting the mineralized/altered rock mass more directly.

Section 10 — Stable isotopes & tracers (cells 26-29)

δ¹⁸O vs δ²H (Craig plot, cell 26): samples fall close to the global/local meteoric water lines rather than shifted to the lower-right — meaning these waters are essentially unmodified meteoric water (rain/snowmelt) with no significant evaporative enrichment before recharge. That's typical of steep, high-elevation catchments with quick infiltration.

δ¹⁸O vs elevation (cell 27): a negative slope ($\approx -0.53\text{‰}$ per 100 m) is the classic isotope-altitude effect, and it lets you back-calculate an apparent recharge elevation for each sample — a way of tracing which part of the catchment (and therefore which flow path through the fractured bedrock/mineralized zone) fed a given well or spring.

Tritium (cell 28): all site medians sit well above the ~0.5 TU modern/sub-modern threshold, so this is a modern, actively-recharging system rather than old, stagnant mine-pool water — though bedrock wells show a wider spread toward lower values, hinting at a minor older-water component mixed in at depth.

Noble gases, He vs Ne (cell 29): bedrock wells show distinctly elevated He (up to ~78 ccSTP/g) relative to springs and streambank wells (~35–40 ccSTP/g). Excess He in groundwater typically accumulates from radiogenic ⁴He produced by U/Th decay in the host rock over time, so this points to bedrock groundwater having a longer subsurface residence time/deeper flow path than the shallow, fast-flushing spring and streambank waters.

Section 11 — Metal-loading regression & contribution heatmap (cells 31-32)

The linear loading rates (mg/s per km of stream) rank **Fe (40.3) > Mn (17.1) > Al (14.6) > Zn (8.2) > Cu (0.54 µg/s equivalent) > Cd (0.033)** — basically tracking each metal's relative abundance in the sulfide/oxide assemblage and its solution behavior rather than toxicity. The contribution-percent heatmap makes the source unambiguous: station **GG-0423** alone accounts for 74–94% of the total load gain for every one of the six metals, confirming a single dominant discharge zone rather than diffuse loading along the whole reach. Several downstream stations show small **negative** contributions, especially for Fe (e.g., -24% at GG-1311) — this is natural attenuation: as the acidic plume mixes with more neutral stream water downstream, dissolved Fe²⁺ oxidizes and precipitates as Fe(III) oxyhydroxides (goethite/schwertmannite-type coatings, the orange "yellow boy" staining typical of AMD streams), which also scavenges some Al, Cu, and Cd onto their surfaces via adsorption/co-precipitation.

Section 12 — Zn load vs. streamflow (cell 33)

Cumulative Zn load tracks close to a 1:1 line against cumulative streamflow, meaning Zn behaves essentially **conservatively** downstream of its source — unlike Fe, it isn't being pulled out of solution by precipitation at the pH range in this stream. This Fe-attenuates/Zn-persists contrast is a well-documented AMD pattern and matters for remediation prioritization: iron's visual/mass signature fades downstream, but Zn (along with Cd, riding along as its lattice substituent) keeps traveling and is usually the metal of greatest downstream ecological concern in these systems.

Note: the two prior tool-produced figures (heatmap and Piper diagram) reflect the corrected versions from earlier in this conversation, not the originally buggy ones.